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(54) EXCAVATOR THUMB ACCESSORY

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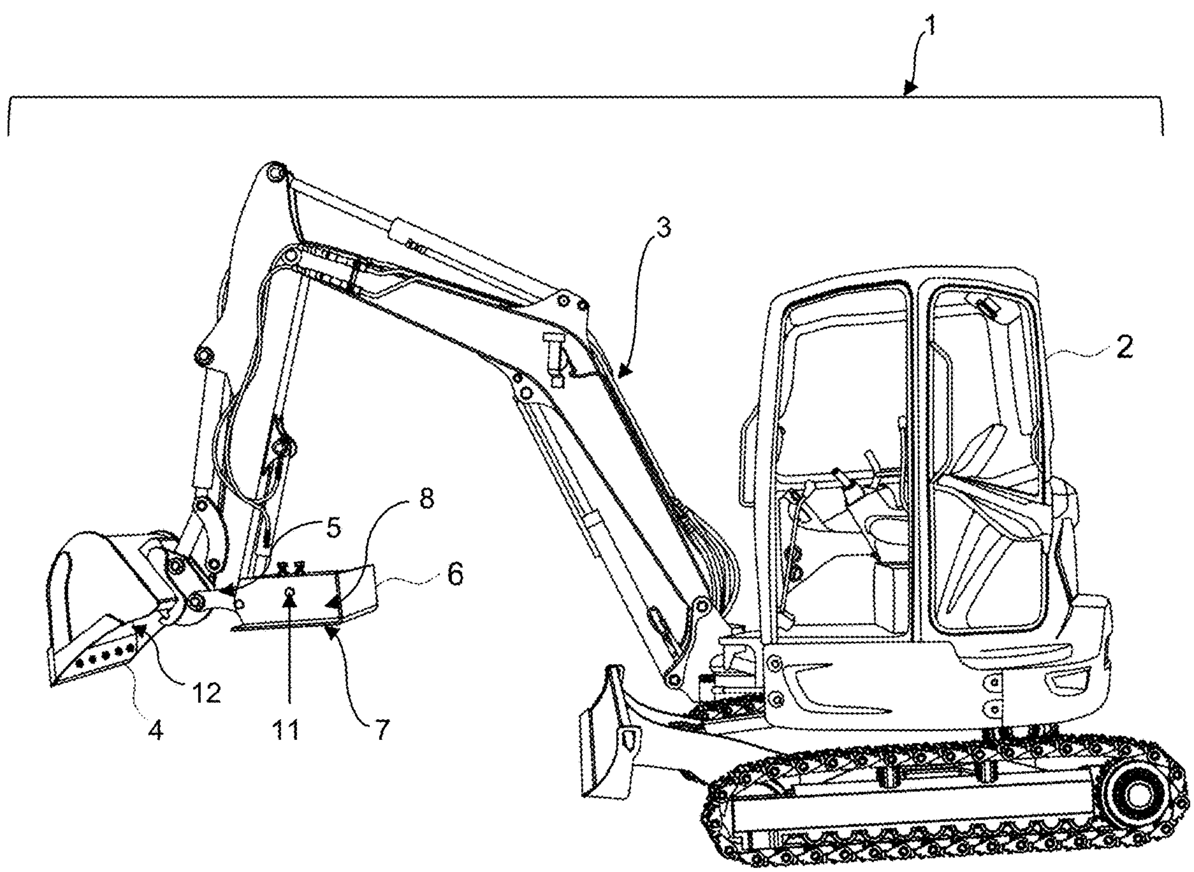
CPC *E02F 3/402* (2013.01)

(57) **ABSTRACT**

An accessory for an excavator is presented wherein the excavator has a bucket with an opening and a moveable thumb on the opposite of the bucket opening. The accessory comprises a plate sized to cover at least a portion of the opening of the bucket and a socket for receiving the thumb to attach the plate to the thumb thereby allowing said plate to move with the thumb. Operation of the thumb causes the plate to cover or uncover the portion of the opening of the bucket.



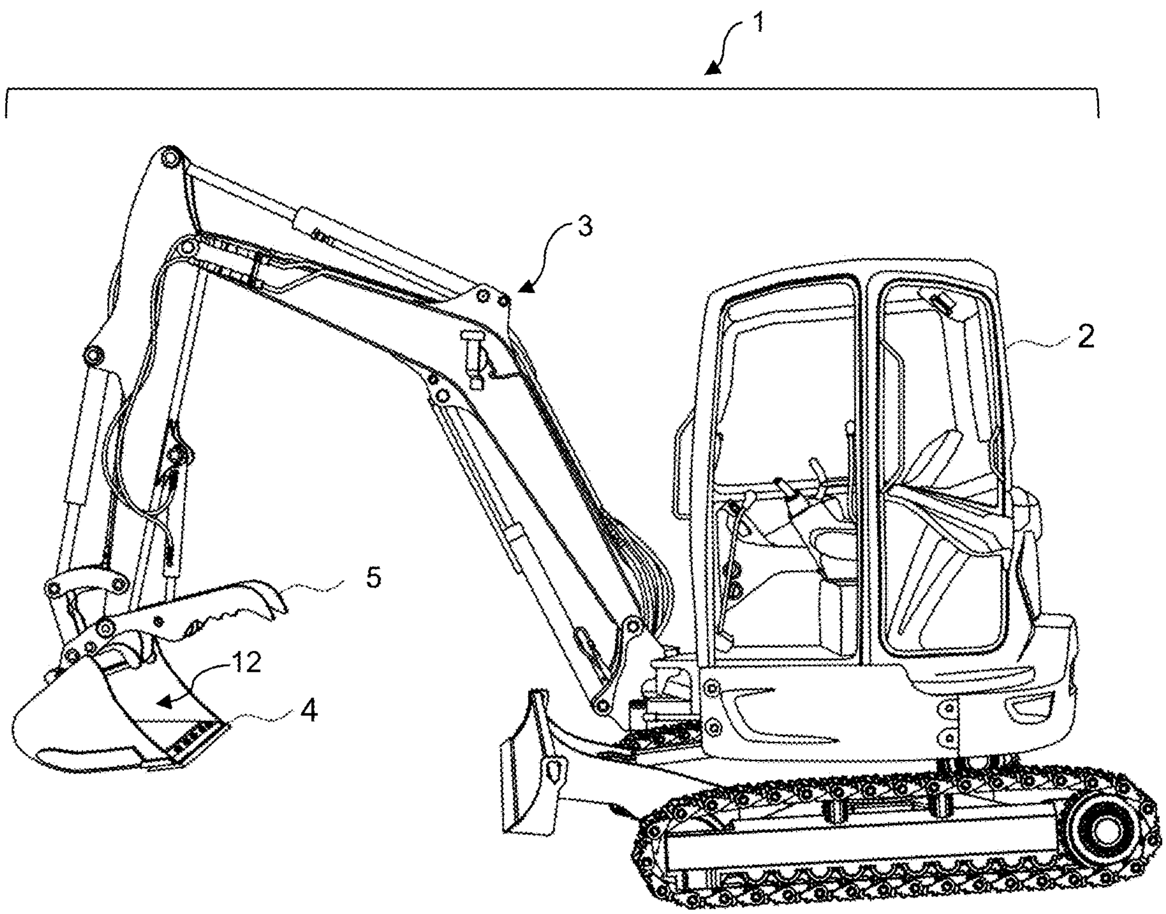


FIG. 1

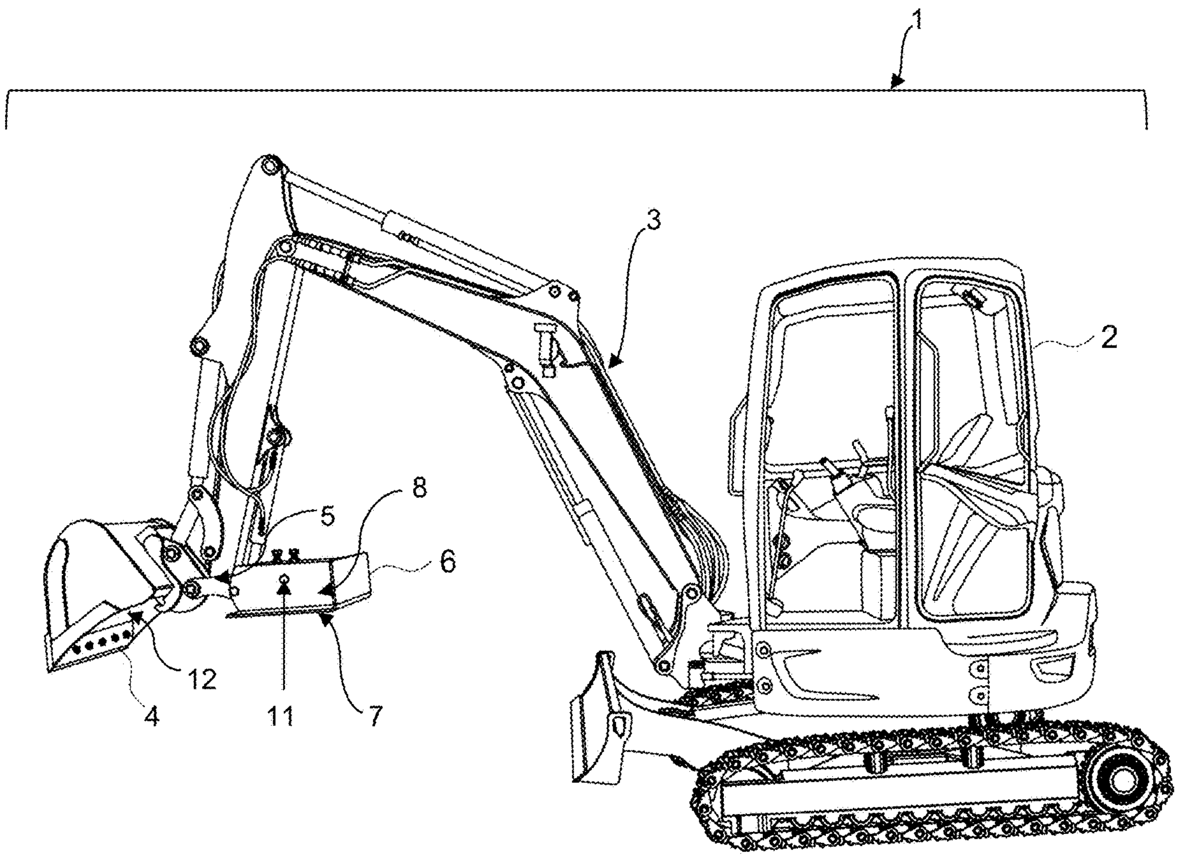


FIG. 2

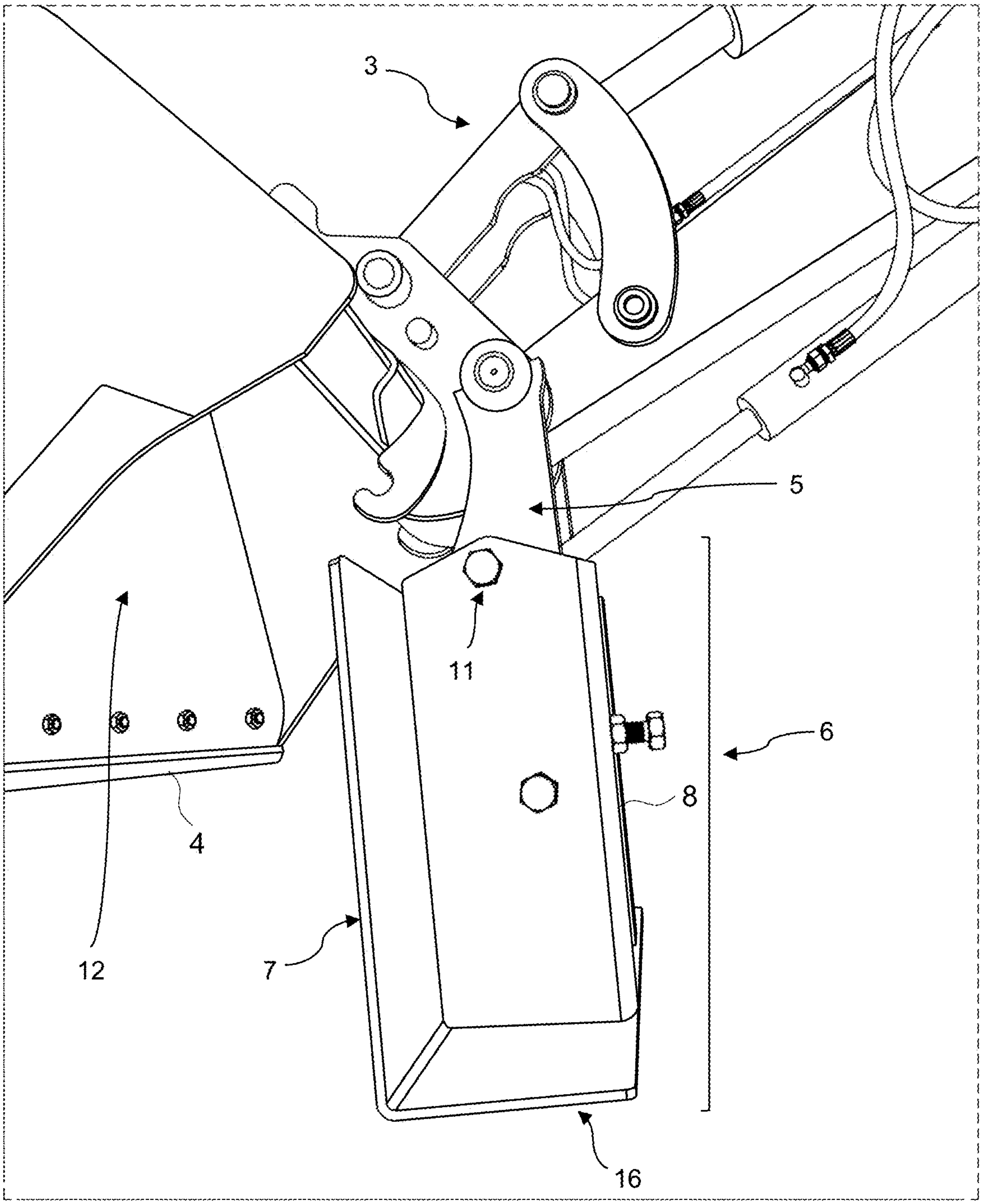


FIG. 3

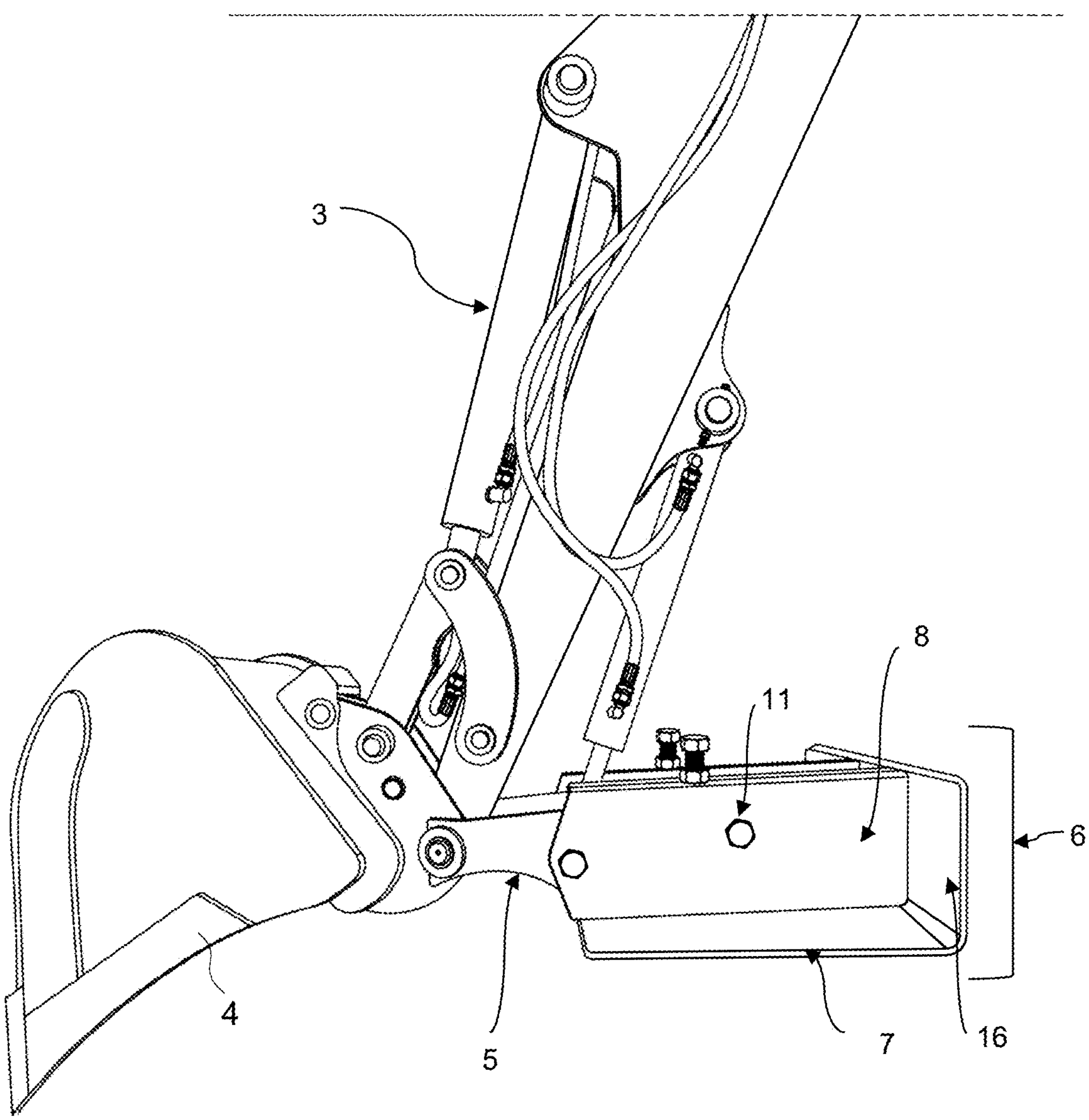


FIG. 4

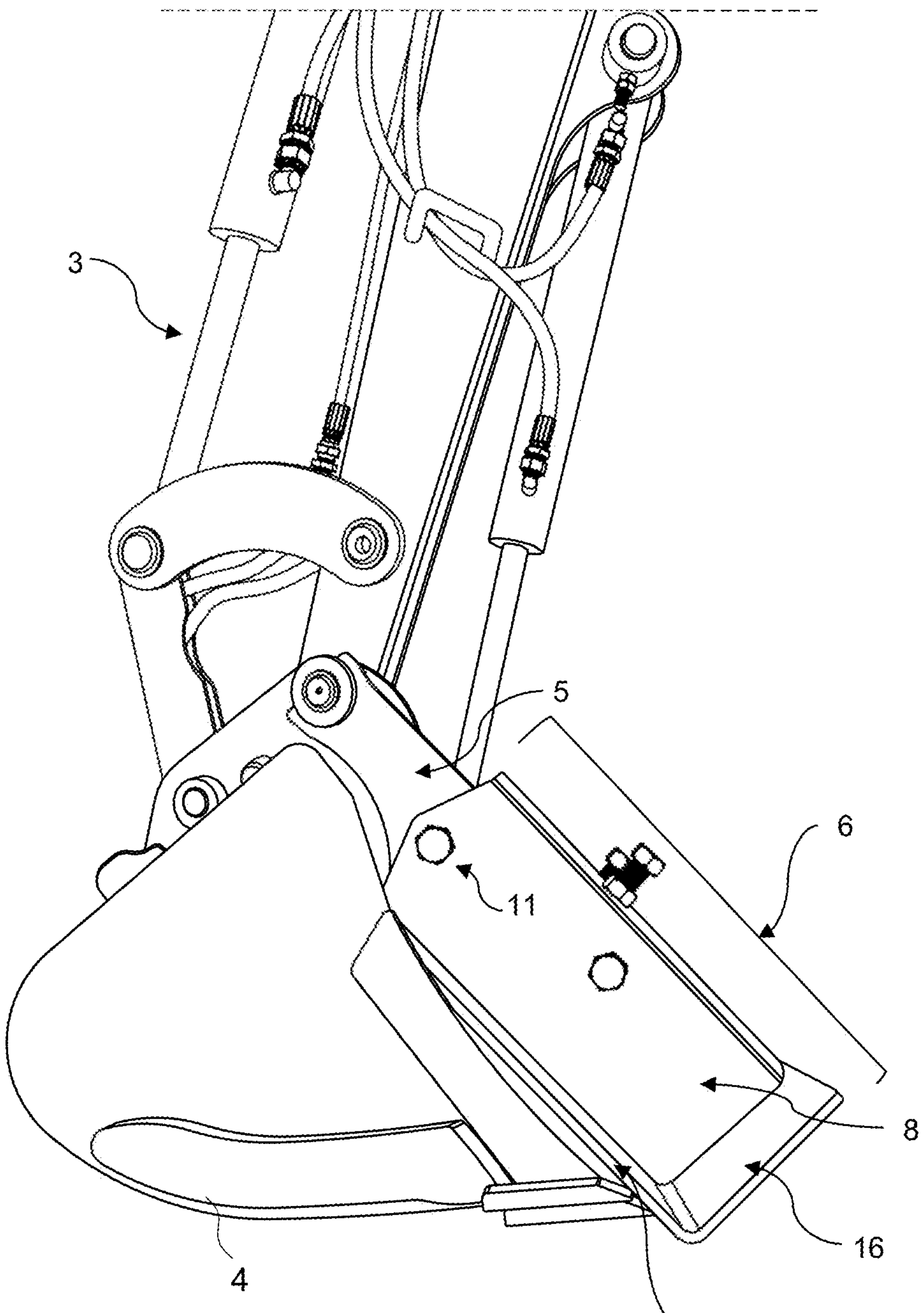


FIG. 5

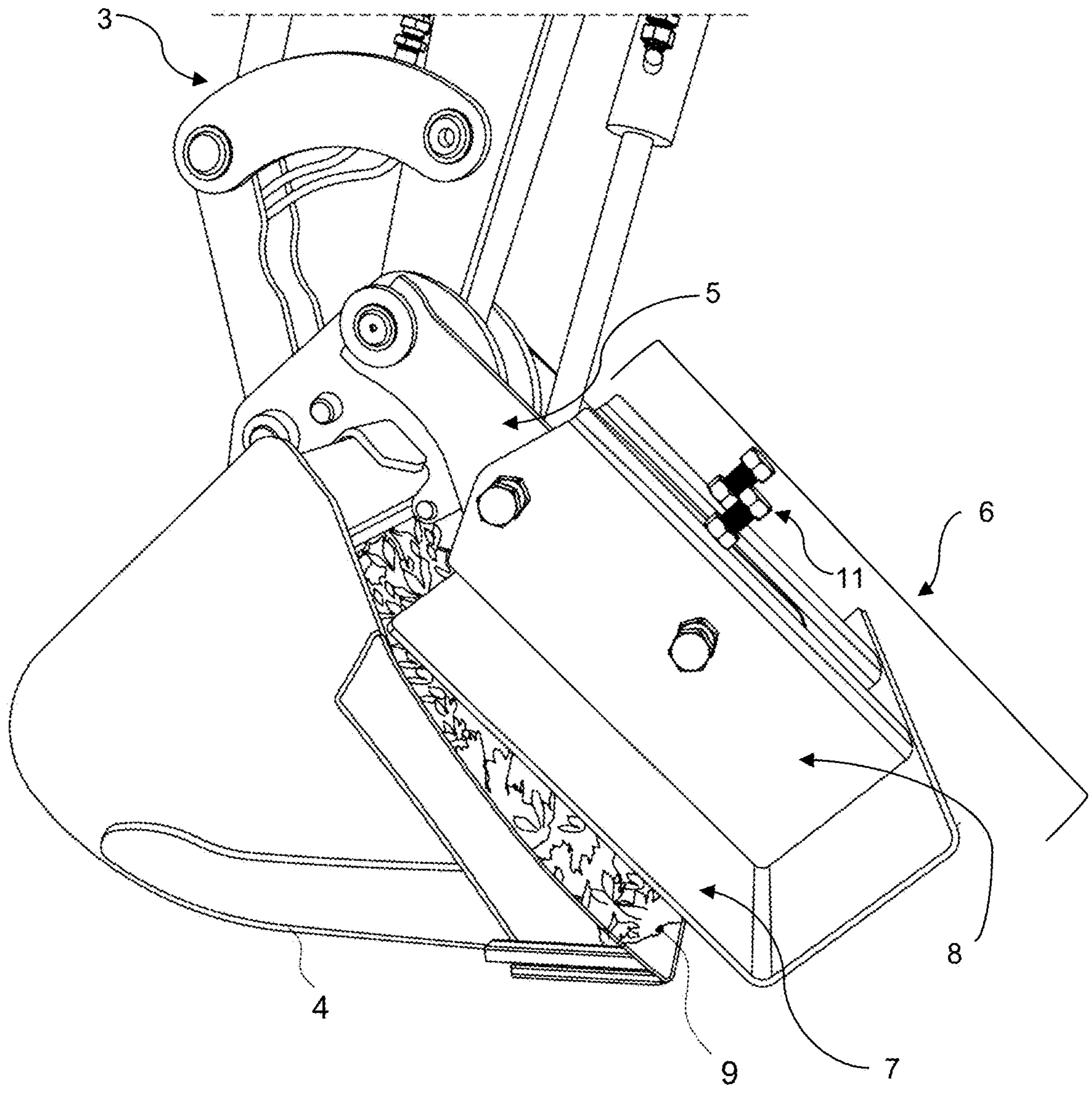


FIG. 6

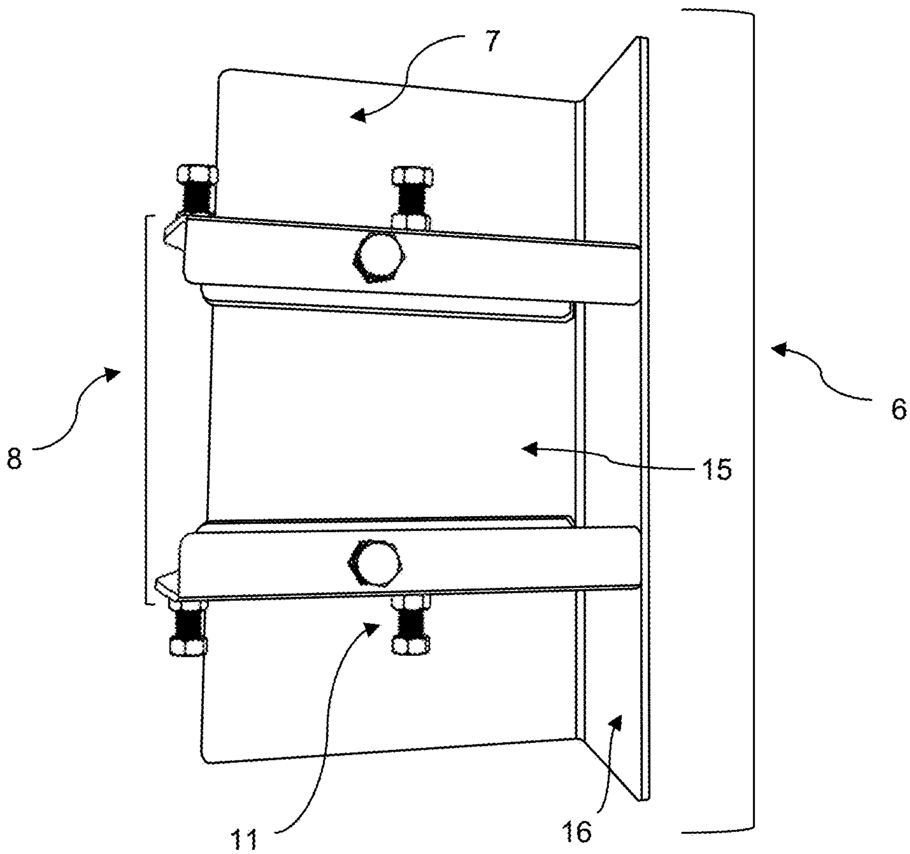


FIG. 7A

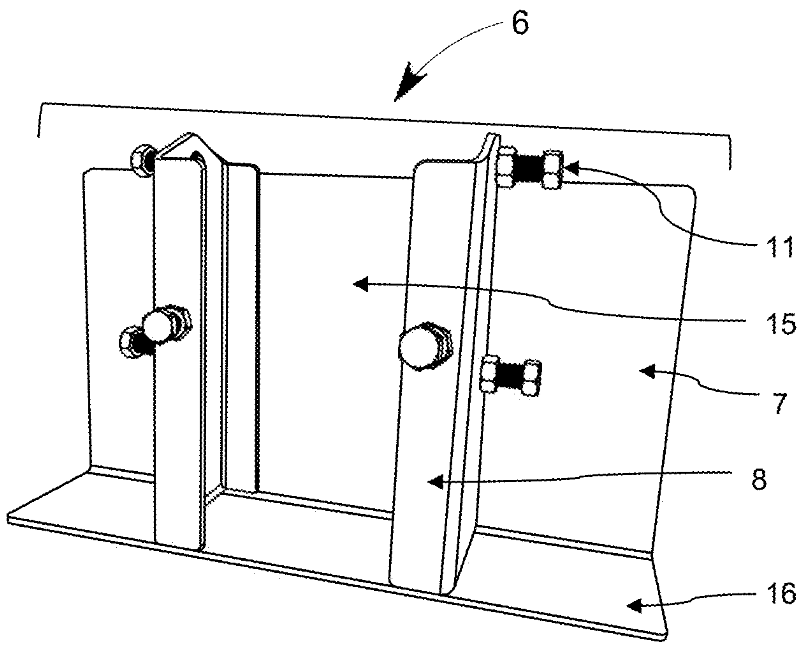


FIG. 7B

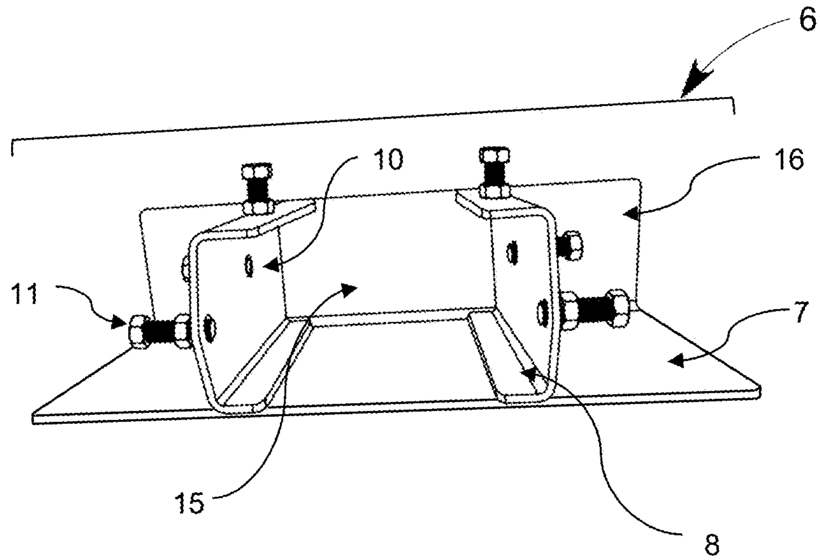


FIG. 7C

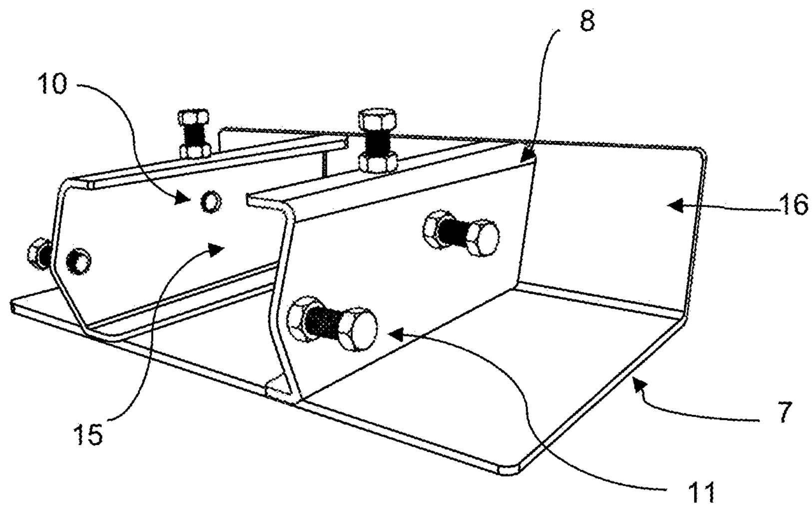


FIG. 7D

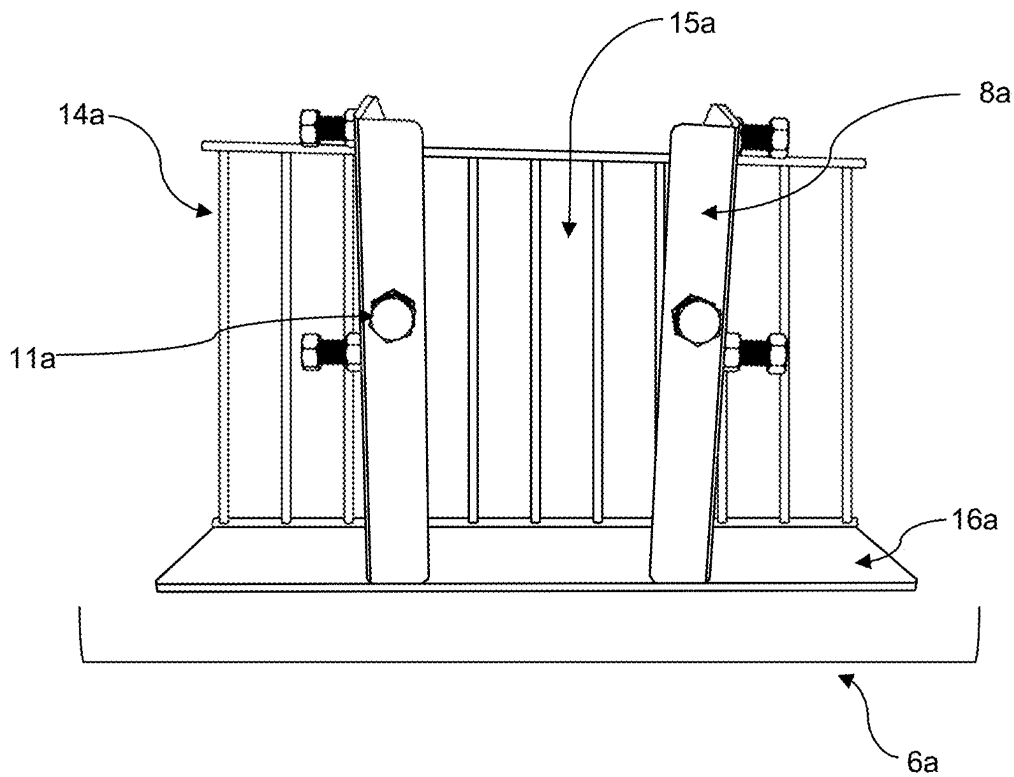


FIG. 8

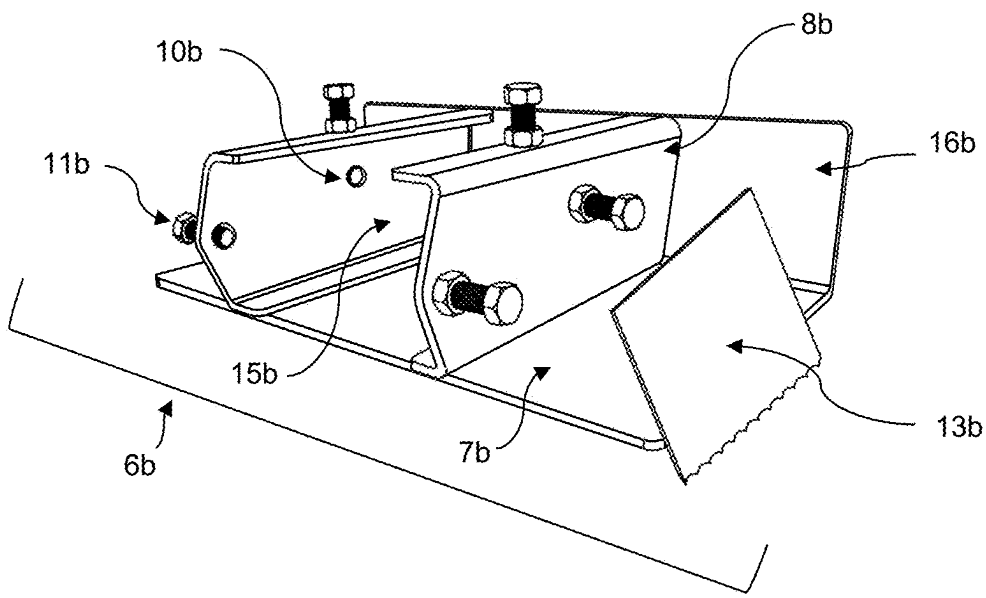


FIG. 9

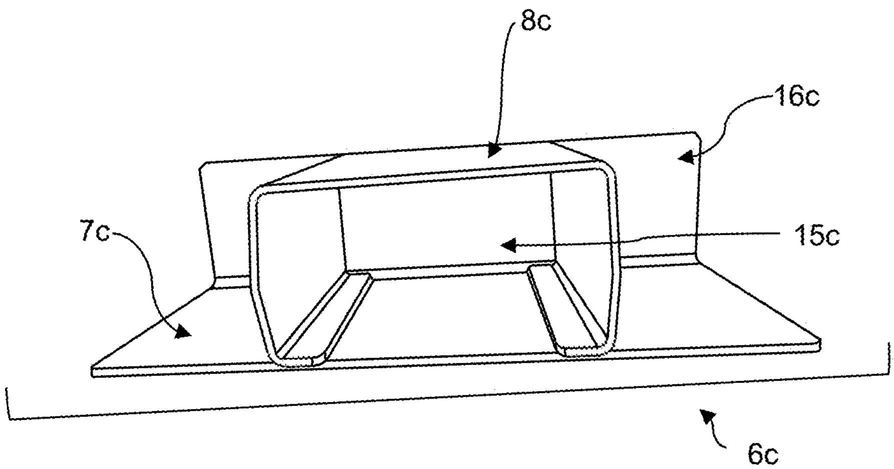


FIG. 10

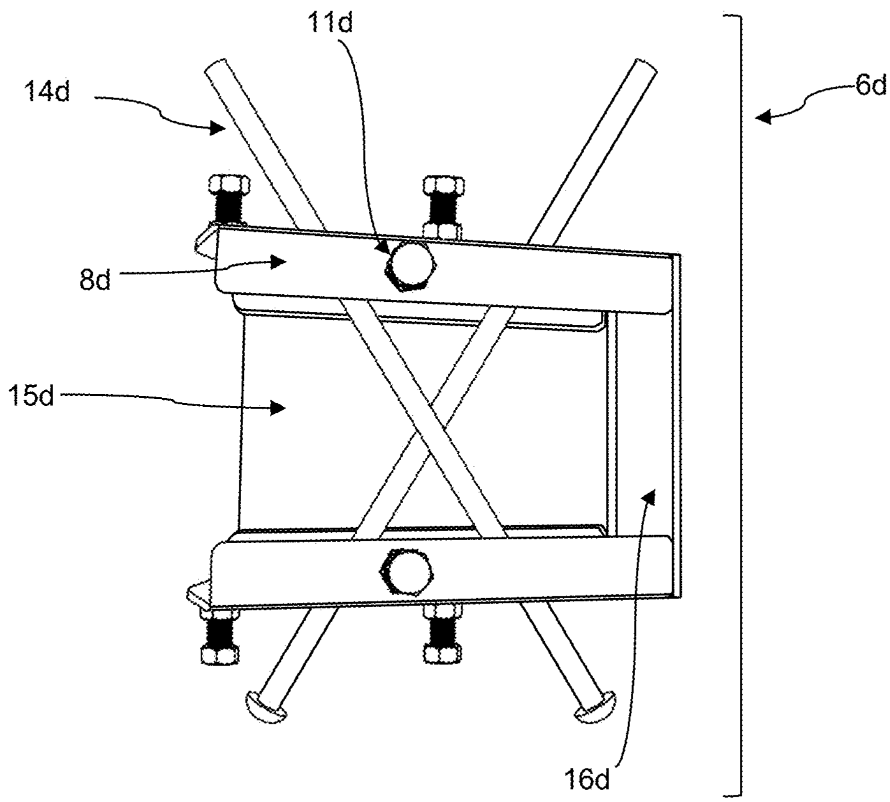


FIG. 11

EXCAVATOR THUMB ACCESSORY

CROSS-REFERENCE

[0001] This application claims the benefit of U.S. Provisional Application No. 63/563,169 filed Mar. 8, 2024, which application is incorporated herein by reference.

BACKGROUND

[0002] What is presented are accessory devices for excavators and mini excavators. Excavators are heavy construction equipment used for digging, lifting and moving materials. However, there are problems with current excavators. The problems include disturbing the ground and difficulty keeping materials in the excavator bucket. When using a grading/ditching bucket, small amounts of material are challenging to get into and keep in the bucket. Also, when trying to remove brush, it is challenging to dig out roots. Another challenge when digging swale ditches is to get the end of the scoop into the bucket cleanly without leaving material. Most operators rely on hand shoveling the remaining material into the bucket.

SUMMARY

[0003] What is presented is an accessory for an excavator or mini excavator wherein the excavator has a bucket with an opening and a moveable thumb on the opposite of the bucket opening. The accessory comprises a plate sized to cover at least a portion of the opening of the bucket and a socket for receiving the thumb to attach the plate to the thumb thereby allowing the plate to move with the thumb. Operation of the thumb causes the plate to cover or uncover the portion of the opening of the bucket. The accessory is constructed of one of: steel, iron, and plastic. The plate can be of any shape such as cross shaped, rectangular, square, round, and triangular. The plate may be larger than, smaller than, or equal in size to the excavator bucket but larger than the thumb. In various embodiments, the plate is solid, mesh, barred, or crossed. In some embodiments, the plate has a lip on a downward edge side that is bent upward into a 90° angle. In some embodiments, the socket can be a variety of configurations such as two bent u-shaped openings, a rectangular opening, and a square opening.

[0004] In various embodiments, the socket has holes for receiving bolts or screws. In various embodiments, the socket is attached to the plate by one of: wielding, bolting, and screws. In various embodiments, a cutting edge is attached perpendicular to a side of said plate and the cutting edge extends toward the bucket when the accessory is attached to the thumb, such that said cutting edge is within the bucket when the plate covers said portion of the opening of the bucket. In other embodiments, the cutting edge is attached perpendicular to a side of said plate and the cutting edge extends toward the bucket when the accessory is attached to the thumb, such that said cutting edge is outside the bucket when the plate covers said portion of the opening of the bucket.

[0005] Those skilled in the art will realize that this invention is capable of embodiments that are different from those shown and that details of the apparatus and methods can be changed in various manners without departing from the scope of this invention. Accordingly, the drawings and descriptions are to be regarded as including such equivalent embodiments as do not depart from the spirit and scope of this invention.

BRIEF DESCRIPTION OF DRAWINGS

[0006] For a more complete understanding and appreciation of this invention, and its many advantages, reference will be made to the following detailed description taken in conjunction with the accompanying drawings.

[0007] FIG. **1** is an external view of an excavator with the bucket and thumb identified;

[0008] FIG. **2** is an external view of an excavator with the accessory attached to the thumb;

[0009] FIG. **3** is a diagram illustrating the placement of the accessory on the thumb;

[0010] FIG. **4**. is a diagram illustrating the positioning of the accessory when the thumb is open;

[0011] FIG. **5** is a diagram illustrating the positioning of the accessory when the thumb is closed;

[0012] FIG. **6**. is a diagram showing the accessory operating with debris in the bucket;

[0013] FIG. **7**A-D are diagrams illustrating different perspectives of the accessory showing the plate and socket;

[0014] FIG. **8** is a diagram showing another embodiment of the accessory where the plate is a barred plate;

[0015] FIG. **9** is a diagram illustrating another embodiment of the accessory where the plate comprises a cutting edge;

[0016] FIG. **10** is a diagram illustrating another embodiment of the accessory with a different configuration of socket;

[0017] FIG. **11** is a diagram showing another embodiment of the accessory in which the plate is a crossed plate.

DETAILED DESCRIPTION

[0018] Referring to the drawings, some of the reference numerals are used to designate the same or corresponding parts through several of the embodiments and figures shown and described. Corresponding parts are denoted in different embodiments with the addition of lowercase letters. Variations of corresponding parts in form or function that are depicted in the figures are described. It will be understood that variations in the embodiments can generally be interchanged without deviating from the invention.

[0019] FIG. **1** illustrates a side view of an excavator **1**. The excavator **1** comprises a cab **2**, an arm **3**, a bucket **4**, and a thumb **5**. The bucket **4** is attached to the end of the arm **3** opposite the cab **2**. The bucket **4** has an opening **12** to load items into the bucket **4** for transportation or removal. The thumb **5** is pivotally attached opposite the bucket **4** opening **12**. The thumb **5** can be in the open position as seen in FIG. **1** or, as shown in FIG. **5**, in the closed position, where the thumb **5** is parallel to the bucket **4** opening **12**. The purpose of the thumb **5** is to help secure and hold items in the bucket **4** to keep them in place during transportation. However, because of the thumb's **5** size and shape, items can fall out of the bucket **4** opening **12**. The proposed device helps to overcome these limitations by creating a covering over the bucket **4** opening **12**.

[0020] FIG. **2** depicts the excavator **1** with the thumb **5** in an open position, moved away from the bucket **4** opening **12**. Attached to the thumb **5** is the accessory **6**. The accessory **6** is comprised of a plate **7** and a socket **8**. The socket **8** is for attaching the plate **7** to the excavator **1** thumb **5**. As shown in the figure, the socket **8** receives the thumb **5** and is secured to the thumb **5** with bolts **11** to allow the plate **7** to move with the thumb **5**. The operation of the thumb **5** causes the plate **7** to cover or uncover said portion of the opening **12** of the bucket **4**. In the open position, the thumb **5** and accessory **6** are positioned away from the bucket **4** opening **12**, uncovering the opening **12** of the bucket **4** thereby allowing items to be scooped into the bucket **4**. When the thumb **5** is in the closed position, the accessory **6** covers a portion of the bucket **4** opening **12**, preventing materials from falling out of the bucket **4**. The accessory **6** may be made of steel, iron, plastic, or any other sturdy material that can withstand the normal operation of the excavator **1**. The plate **7** is sized to cover at least a portion of the opening **12** of the bucket **4** but it will be understood that different sized plates **7** may be used depending on the requirements for the accessory **6**. The plate **7** could be larger or smaller than the opening **12** of the bucket **4**.

[0021] FIG. **3** shows an enlarged view of the accessory **6**. In this embodiment, the plate **7** and the socket **8** are secured to the thumb **5** with bolts **11**.

[0022] FIG. **4** illustrates the thumb **5** in the open position. Mounted on the thumb **5** is the accessory **6**. In the open position, the thumb **5** and accessory **6** are positioned away from the bucket **4** opening **12**, allowing the bucket **4** to be used to dig and scoop up debris **9** and other materials.

[0023] FIG. **5** shows the thumb **5** and accessory **6** in the closed position. In the closed position, the accessory **6** covers at least a portion of the bucket **4** opening **12**. In creating a surface over the bucket **4** opening **12**, the accessory **6** can be used to hold materials in the bucket **4**, pinch sod or brush, sift through the debris **9**, or cut off the materials in the bucket.

[0024] FIG. **6** shows the bucket **4** containing debris **9**. FIG. **6** also shows the thumb **5** and accessory **6** approaching the closed position.

[0025] The excavator thumb accessory **6** utilizes the plate **7** to create a surface over the bucket **4** of the excavator **1**, allowing material and debris **9** to be trapped in the bucket **4**. Additionally, the plate **7** allows the user to pinch brush to pluck out bushes, minimizing the amount of ground disturbance and excavation. Another challenge when digging swale ditches is to get the end of the scoop into the bucket **4** cleanly without leaving material or debris **9** behind. This accessory **6**, allows the user to curl material and debris **9** into the bucket **4** and pinch off sod thereby cutting the sod or leaving a more precise and clean cut. Most operators rely on hand shoveling the remaining material and debris **9** into the bucket **4**. This accessory **6** enables the operator to stay on the excavator **1** and not need to perform manual labor, there by saving time, money, and reducing physical exertion.

[0026] FIG. **7**A illustrates the accessory **6** not mounted to the excavator **1**. This embodiment of the accessory **6** is comprised of a solid rectangular plate **7** and the socket **8** is formed from two bent-u shapes. The thumb **5** slides between the two bent-u shapes comprising the socket **8**. Though the accessory **6** may be secured to the thumb **5** in various ways, including using screws or welding, the accessory **6** in this embodiment is secured by bolting the socket **8** to the thumb **5**. Holes **10** are places along the socket **8** to allow the bolts **11** or screws or other attachment systems to secure the accessory **6** to the thumb **5**. The plate includes a lip **16** on a downward edge side that is bent upward into a 90° angle. The purpose of the lip **16** is to more securely attach the thumb **5** to the socket **8**.

[0027] FIG. **7**B details the accessory **6** as shown in FIG. **7**A rotated 90 degrees to more clearly show the elements of the accessory **6**.

[0028] FIG. **7**C illustrates the accessory **6** from another perspective. From this angle, the opening **15** of the socket **8** is depicted. Additionally, FIG. **7**C shows holes **10** in the socket **8** so that the accessory **6** can be bolted to the thumb **5**.

[0029] FIG. **7**D shows the placement of bolts **11** and holes **10** along the socket **8**.

[0030] FIG. **8** shows another embodiment of the accessory **6***a*. In this embodiment, the plate **7***a* is comprised of bars **14***a*. The bars **14***a* allow smaller materials to fall out of the bucket **4***a* while keeping larger materials and debris **9***a* trapped in the bucket **4***a*. The socket **8***a* is comprised of two bent-u shapes with holes **10***a* placed to allow the accessory **6***a* to be attached to the thumb **5***a* with bolts **11***a*. The space between the bars **14***a* can be varied to screen different sizes of materials and debris **9***a*. In fact, the bars **14***a* could be replaced with or include a mesh material (not shown) for sifting of finer debris **9***a*.

[0031] FIG. **9** shows another embodiment of the accessory **6***b*. In this embodiment, a cutting edge **13***b* is attached to the side of the plate **7***b*. The cutting edge **13***b* can be used to cut materials in the bucket **4***b* when the thumb **5***b* is moved to the closed position. The cutting edge **13***b* enters the bucket **4***b* when the thumb **5***b* is in the closed position. The cutting edge **13***b* reduces the size of the debris **9***b* so that it is contained within the bucket **4***b* which makes the debris **9***b* easier to transport by the excavator **1***b*. Furthermore, the cutting edge **13***b* allows the bucket **4***b* to cut debris **9***b* to more easily be scooped within the bucket **4***b*. When the thumb **5***b* is in the closed position, the cutting edge **13***b* could be on the outside or the inside of the bucket **4***b*.

[0032] FIG. **10** shows another embodiment of the accessory **6***c*. In this embodiment, the plate **7***c* is a solid rectangle. Moreover, the socket **8***c* is a rectangular shape allowing the thumb **5***c* to slide in the opening **15***c*. The socket **8***c* in this embodiment does not have holes **10***c* to allow for bolting, but instead the socket **8***c* can be attached by welding. Other socket **8***c* configurations are also possible such as a square or other shapes.

[0033] FIG. **11** illustrates another embodiment of the plate **7***d*. This accessory **6***d* has a plate **7***d* that is comprised of two crossed bars **14***d*. Here, the socket **8***d* is comprised of two bent-u shapes with holes **10***d* placed to allow the accessory **6***d* to be attached to the thumb **5***d*. The thumb **5***d* would slide though the opening **15***d* of the socket **8***d*. Other shapes of plates **7***d* are possible such as square, round, and triangular so long as the plate **7***d* covers a portion of the opening **12***d* of the bucket **4***d*.

[0034] This invention has been described with reference to several preferred embodiments. Many modifications and alterations will occur to others upon reading and understanding the preceding specification. It is intended that the invention be construed as including all such alterations and modifications in so far as they come within the scope of the appended claims or the equivalents of these claims.

What is claimed is:

1. An accessory for an excavator or mini excavator comprising:

wherein excavator has a bucket with an opening and a moveable thumb on the opposite of the bucket opening;

a plate sized to cover at least a portion of the opening of the bucket; and

a socket for receiving the thumb to attach said plate to the thumb thereby allowing said plate to move with the thumb wherein operation of the thumb causes said plate to cover or uncover said portion of the opening of the bucket.

2. The accessory of claim **1** wherein said plate is constructed of one of: steel, iron, and plastic.

3. The accessory of claim **1** wherein said plate is one of: cross shaped, rectangular, square, round, and triangular.

4. The accessory of claim **1** wherein said plate is larger than, smaller than, or equal in size to the excavator bucket but larger than the thumb.

5. The accessory of claim **1** wherein said plate is solid, mesh, barred, or crossed.

6. The accessory of claim **1** wherein said plate has a lip on a downward edge side that is bent upward into a 90° angle.

7. The accessory of claim **1** wherein said socket comprises one of: two bent u-shaped openings, a rectangular opening, and a square opening.

8. The accessory of claim **1** wherein said socket has holes for receiving bolts or screws.

9. The accessory of claim **1** wherein said socket is attached to said plate by one of: wielding, bolting, and screws.

10. The accessory of claim **1** wherein a cutting edge is attached perpendicular to a side of said plate and said cutting edge extends toward the bucket when the accessory is attached to the thumb, such that said cutting edge is within the bucket when the plate covers said portion of the opening of the bucket.

11. The accessory of claim **1** wherein a cutting edge is attached perpendicular to a side of said plate and said cutting edge extends toward the bucket when the accessory is attached to the thumb, such that said cutting edge is outside the bucket when the plate covers said portion of the opening of the bucket.

* * * * *